Mukul Kumar

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PROFILE

Computer Science undergraduate (CGPA 8.34) and Oracle Certified Java SE 17 Developer who builds REST APIs in Java with Spring Boot and in Node.js. Deployed Dockerized backend microservices to AWS EC2 during a backend and DevOps internship, and built a vector database from scratch in C++ with a RAG pipeline on top. Codes with GitHub Copilot and Cursor. Seeking a Software Development Engineer internship in Java, cloud and AI-driven backend engineering.

EDUCATION

SRM University–AP, Amaravati 2023 – 2027
B.Tech. in Computer Science and Engineering | CGPA: 8.34 / 10

TECHNICAL SKILLS

- **Languages:** Java, C++, Python, TypeScript, JavaScript, SQL
- **Backend and Cloud:** Spring Boot, Node.js, NestJS, FastAPI, REST APIs, JSON, JWT, RBAC, AWS (EC2), Docker, Nginx, CI/CD
- **Data and AI:** PostgreSQL, MongoDB, Redis, RAG, LLM (Ollama), Vector Search (HNSW), Generative AI, TensorFlow
- **Core:** Data Structures and Algorithms, Object-Oriented Programming, Problem Solving, System Design, DBMS, Operating Systems, AI-driven development
- **Tools:** Git, Linux, GitHub Copilot, Cursor, Unit Testing

EXPERIENCE

Backend and DevOps Intern, Ewhent Technologies Jun 2025 – Aug 2025
Node.js, Docker, AWS EC2, MongoDB, Redis, Nginx, REST APIs

- Designed role-based access control and 10+ REST APIs in Node.js and MongoDB for authentication and authorization.
- Containerized and deployed 3+ backend microservices to AWS EC2 with Docker and Nginx, making releases from development to production more consistent.
- Automated infrastructure workflows for analytics pipelines and ML inference, cutting manual deployment steps by 30%; added Redis caching and session sync.

PROJECTS

Offline UPI Mesh Network Simulator GitHub
Java, Spring Boot, H2, REST APIs, RSA-OAEP, AES-256-GCM, SHA-256

- Built an offline payment settlement system in Java and Spring Boot with REST APIs that forward encrypted transactions across a mesh-network routing design.
- Secured payloads with RSA-OAEP and AES-256-GCM and added concurrency-safe idempotency handling.
- Structured the code as modular, testable services using clean architecture, with sub-200ms response times.

Meai: Vector Database Built from Scratch GitHub
C++, HNSW, KD-Tree, REST APIs, Ollama, RAG

- Built a vector database in C++ with HNSW, KD-Tree and brute-force search behind a REST API for CRUD and search.
- Connected a RAG pipeline that pairs a local LLM (Ollama) with the HNSW index for context-grounded question answering.
- Benchmarked cosine, Euclidean and Manhattan metrics on 768-dimension embeddings, with O(log N) HNSW search.

KrishiAI: AI-Powered Smart Agriculture Platform GitHub
Python, Flask, TensorFlow, CNN, XGBoost, SQLAlchemy, JavaScript

- Trained CNN plant-disease classifiers reaching 87%+ accuracy across 10+ crop categories, served through real-time Flask REST API inference.
- Developed XGBoost crop recommendation and fertilizer optimization models, with multilingual support and live weather API integration.
- Built the JavaScript, HTML and CSS frontend on Flask REST APIs and unit tested both client and server layers.

CERTIFICATIONS AND ACHIEVEMENTS

- Oracle Certified Professional: Java SE 17 Developer (2026); SAP Generative AI Developer (2026).

LEADERSHIP

- **Co-Head, Technology, GDG SRM-AP:** led technical workshops, coding sessions and community events for 100+ students.
- **Captain, School Football Team:** directed team strategy and coordination in inter-school tournaments.